

## Differences in postprandial inflammatory responses to a 'modern' v. traditional meat meal: a preliminary study.

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A low-grade inflammatory response ('metaflammation') has been found to be associated with certain chronic diseases. Proposed inducers of this have been aspects of the modern lifestyle, including newly introduced foods. Plasma TAG, and the inflammatory cytokines C-reactive protein (CRP), TNF-alpha and IL-6 were compared in a randomised, cross-over trial using ten healthy subjects before and after eating 100 g of kangaroo, or a 'new' form of hybridised beef (wagyu) separated by about 1 week. Postprandial levels for 1 and 2 h of TAG, IL-6 and TNF-alpha were significantly higher after eating wagyu compared with kangaroo ( $P = 0.002$  for TAG at 1 h,  $P < 0.001$  at 2 h;  $P < 0.001$  for IL-6 and TNF-alpha at 1 and 2 h). CRP was significantly higher 1 h postprandially after wagyu ( $P = 0.011$ ) and non-significantly higher 2 h postprandially ( $P = 0.090$ ). We conclude that the metaflammatory reaction to ingestion of a 'new' form of hybridised beef (wagyu) is indicative of a low-grade, systemic, immune reaction when compared with lean game meat (kangaroo). Further studies using isoenergetic intake and isolating fatty acid components of meats are proposed.